

Homatics: Delivering a one-stop smart TV solution for quality entertainment and home security

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ABOUT HOMATICS



About Homatics

To deliver a one-stop solution that aggregates different types of streaming services, smart home features. Homatics leverages Cloud to simplify integration and simplify maintenance work.

Homatics is a new international smart solution brand providing end-to-end solutions for seamless audio and video entertainment experiences, as well as smart home services across living rooms and multi-rooms based on the Google ecosystem. In 2019, Homatics launched a smart TV solution that combines broadcast TV, over-the-top media streaming, smart home and security services. As of 2021, the company's smart TV solution has over one million users around the world.

reducing deployment
and costs.

Industries: Technology

Location: Singapore | South Africa |
China | America

Google Cloud results

- Simplifies digital rights management for video streaming services with Live Stream API
- Shortens deployment time of video distribution solution for TV operators from months to days
- Reduces software update time by 50% through quick computing environment duplication supported by GKE

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(<https://cloud.google.com/sql/>), [Cloud Load](#)

[Google Cloud \(<https://cloud.google.com/>\)](https://cloud.google.com/)

(<https://cloud.google.com/load-balancing/>),

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(<https://cloud.google.com/storage/>)

[Cloud SQL \(<https://cloud.google.com/sql/>\)](https://cloud.google.com/sql/)

[Cloud Load Balancing](#)

(<https://cloud.google.com/load-balancing/>)

[Cloud Storage \(<https://cloud.google.com/storage/>\)](https://cloud.google.com/storage/)

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[Kubernetes Engine](#)

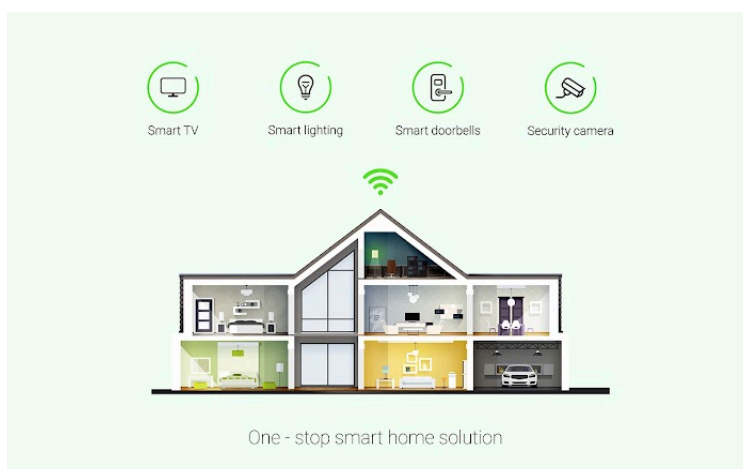
(<https://cloud.google.com/kubernetes-engine/>)

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- Enables quick troubleshooting with detailed log records on Cloud SQL

The growing popularity of smart homes has changed the entertainment experience and home life for many people. Several program options are now available on smart TVs, ranging from broadcast TV channels to over-the-top (OTT) streaming platforms like Netflix and Hulu, while devices like smart doorbells and smart security cameras have greatly enhanced home security and convenience.

However, accessing different content platforms and using multiple smart home appliances is not always a smooth experience. For example, controlling different smart gadgets may require multiple applications, making it complex. Users also notice inconsistencies in video loading speed and features when switching between broadcast TV and online streaming content, which can be frustrating.



Homatics (<https://www.homatics.com/>) is dedicated to delivering streamlined smart home services and immersive TV watching experiences by offering a smart TV solution that seamlessly integrates all types of TV program sources and incorporates smart home features. Homatics' solution not only aggregates digital video broadcasting (DVB) and OTT media services, provides features like show live broadcasting, recording and replay, but is also fully integrated with its self-developed Internet of things (IoT) system.

Developing a one-stop smart TV solution requires a cloud platform that provides powerful streaming video processing and content distribution tools available for all consumers around the world. To that end, Homatics chose Google Cloud (<https://cloud.google.com/>) because it simplifies the integration between the video transcoding and distribution of OTT content and supports fast content delivery through its extensive global network. On top of that, Homatic's self-developed IoT system is fully integrated with Google Assistant.

Google Cloud meets all the company's needs by providing a full set of high-performance tools to process and distribute video content. The out-of-the-box tools and pay-as-you-go scheme of Google Cloud enable Homatics to deploy its smart TV solution that offers excellent TV viewing experiences and well-integrated smart home features at the lowest deployment time and costs. In July 2019, Homatics launched its smart TV software that currently has more than one million users globally.

A one-stop smart TV solution supporting live streaming with very low delays

To transcode the signals of the OTT live streaming service on Homatics' smart TV solution into encrypted output streams, Homatics employs Live Stream API (<https://cloud.google.com/livestream>), which simplifies the setup of digital rights management (DRM), a method of preventing unauthorized use and piracy for digital content, by automating the generation and management of encryption keys. With Live Stream API, Homatics is able to quickly set up digital rights management for new live broadcast channels just by defining the basic parameters, instead of going through complicated encryption processes. This has helped greatly reduce the development time of its smart TV solution.

On top of that, Live Stream API enables the Homatics team to realize the traditional TV broadcast features, such as program reservation, recording and time shifting for OTT live streaming. By using Live Stream API to process video signals and Cloud Storage (<https://cloud.google.com/storage>) to store the log files of smart home appliances, the company is able to offer IoT capabilities like real-time device status updates and home camera live image displays on smart TVs. As a result, Homatics has successfully streamlined the viewing experience of both DVB and OTT live content, while merging smart security and smart home services on its smart TV solution.

Fully integrating DVB, OTT and IoT services means that Homatics can deliver high-quality entertainment and all-in-one smart home experiences, especially in terms of security. Users of its smart TV solution can easily switch between TV and smart home features by connecting the system with home cameras, smart doorbells and multiple smart sensors. For example, users can check the live image captured by home surveillance cameras and receive alerts any time in the middle of a TV show, which allows them to rest assured that they won't overlook any security risks while watching TV.

For data transferring, Homatics leverages Cloud CDN (<https://cloud.google.com/cdn>) to deliver HTTP live streaming (HLS) content globally at a faster speed. Compared to the open source content delivery solution that the company previously used, Cloud CDN can realize nearly no delays for HLS and distribute video content 5X faster by responding to requests more efficiently. Furthermore, Homatics uses Cloud Load Balancing (<https://cloud.google.com/load-balancing>) to better allocate incoming traffic to avoid technical issues caused by flow surges.

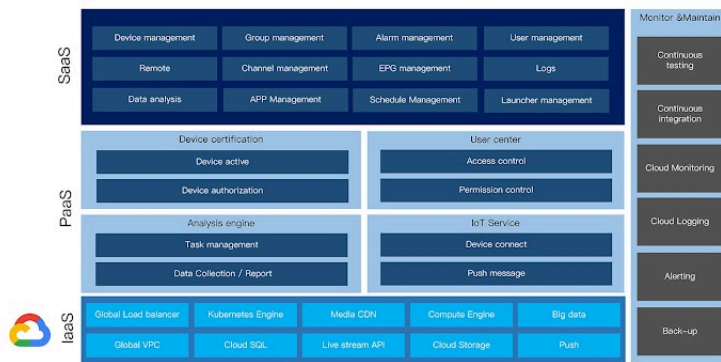
Enhancing development efficiency and reducing deployment costs

One of the greatest advantages of using Google Cloud for Homatics is that it has helped significantly enhance software development and deployment efficiency while minimizing the IT infrastructure costs. With out-of-the-box tools like Live Stream API and Cloud CDN, Homatics is able to deploy its video

distribution system much faster than building one from scratch. The pay-as-you-go scheme of Google Cloud products also enables Homatics to launch the operation of the smart TV software without investing too much in IT infrastructure beforehand.

"Developing and deploying a smart TV solution from scratch in an on-prem environment could easily take more than half a year. With the cloud infrastructure and ready-to-use solutions of Google Cloud, Homatics can deploy our smart TV solution within one week," says James Wen, co-founder of Homatics.

Homatics also keeps improving its IT work process for greater efficiency. For example, the company needs to update its smart TV software every month, and it used to take hours to deploy an update because the microservices that support its smart TV system were not managed in an aligned manner. Service was sometimes disrupted, because Homatics' engineers had to shut down the system to clear the errors that appeared during the implementation of an update. To increase update efficiency, Homatics moved the microservices of its smart TV system to Google Kubernetes Engine (GKE) for centralized management and aligned the setup of its development environment with the operation environment on GKE, which has allowed it to easily deploy software updates through duplication, instead of manually configuring all the settings. This has helped the Homatics' development team reduce the time required for deploying a software update by 50%, and minimize setting errors.



Enabling quick troubleshooting with detailed log records

Another benefit Homatics' engineers enjoy with Google Cloud products is the accelerated troubleshooting process. The team relies on Cloud SQL (<https://cloud.google.com/sql>) to host the databases of its smart TV system and applications. In one instance, the inquiry speed of the databases on Cloud SQL slowed down significantly when the amount of usage went up, which is abnormal because Cloud SQL is highly scalable. By consulting the detailed log records of Cloud SQL, the engineering team quickly identified the setting errors and permanently fixed the issue. Its databases on Cloud SQL have since been running smoothly under any level of usage.

When using Cloud Load Balancing to distribute incoming traffic across its gateways, Homatics' engineers can analyze the performance of each gateway by checking the distribution records of Cloud Load Balancing. Thanks to this feature, the team swiftly pinpointed the issue cause when it noticed that a lot of access requests failed during a stress test of its IoT system.

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*—James Wen, Co-
Founder, Homatics*

Creating broader services and higher business returns

Homatics is looking to expand its use of Google Cloud to enhance its smart home services and expand its businesses centered around home TV screens. The company plans to add new media features like on-demand videos and cloud recording supported by Live Stream API and Cloud Storage, provide even faster content delivery speed by replacing Cloud CDN with Media CDN

(<https://cloud.google.com/media-cdn>), and enhance its smart home security features by leveraging Video AI (<https://cloud.google.com/video-intelligence>) to analyze images captured by smart cameras and detect security risks in real time.

To keep operational costs low and increase its data analytics capabilities, Homatics will move its big data system to BigQuery

(<https://cloud.google.com/bigquery>). It will also insert ads into video content on its smart TV solution to monetize its services by connecting Video Stitcher API (<https://cloud.google.com/video-stitcher>) and Google Ads with Live Stream API.

"Our goal is to deliver a smart TV solution that not only offers the best possible services to end users," says Wen. "Thanks to Google Cloud, we've successfully created streamlined TV entertainment and smart home experiences by fully integrating different media sources and smart systems. We're now looking for new ways we can continue broadening our services with a powerful set of tools."

